Below is a **complete developer functional document** describing the system behavior and architecture based on everything we discussed.
This document **focuses on system functionality, workflows, and modules**, intentionally **excluding database schemas or table definitions**.

Multi-Tenant School Management & Learning System

Developer Functional Specification

1. System Overview

The platform is a **multi-tenant School Management System (SMS)** designed to support schools from **Infant level (ECD)** through **Primary and Secondary education** within a single unified system.

The system combines three major components:

1. **School Administration System**
2. **Learning Management System (LMS)**
3. **Academic Assessment and Reporting Engine**

Each school operates as an **independent tenant** within the platform while being managed centrally by a **Super Administrator**.

The system is designed to support **multiple roles**, each with specific responsibilities and controlled access to system resources.

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2. System Roles and Portals

The system contains several user portals corresponding to different roles within the educational ecosystem.

2.1 Super Administrator Portal

The Super Administrator manages the entire platform across all schools.

Responsibilities include:

- Creating and managing schools
- Configuring platform-level settings

- Managing subscription plans or service packages
- Monitoring system usage across schools
- Viewing platform-wide analytics
- Managing system updates and maintenance
- Managing global configurations such as grading defaults and academic structures

This portal is intended only for **platform owners or system administrators**.

2.2 School Administrator Portal

Each school has a dedicated administrator responsible for managing the school's operations within the platform.

Responsibilities include:

- Managing teachers
- Registering and managing students
- Managing parents
- Creating classes and assigning class teachers
- Creating departments
- Creating and assigning subjects
- Creating academic terms and academic years
- Creating and managing assessments and exams
- Managing timetables
- Publishing report cards
- Monitoring attendance
- Managing learning materials
- Configuring school-specific academic policies

School administrators oversee all academic and administrative activities for their school.

2.3 Head of Department (HOD) Portal

Departments represent subject groupings within the school such as:

- Languages
- Sciences
- Commercial subjects
- Arts

Each department has a **Head of Department** responsible for overseeing the academic performance of that department.

Responsibilities include:

- Monitoring subject teachers within the department
- Reviewing student performance for departmental subjects
- Monitoring subject pass rates
- Reviewing and validating marks entered by teachers
- Providing departmental reports and analytics
- Uploading departmental learning materials
- Supporting teachers in curriculum delivery

HODs function as academic supervisors for subject areas.

2.4 Teacher Portal

Teachers interact with the system for both academic instruction and student evaluation.

Teachers may teach **multiple subjects across multiple classes**.

Responsibilities include:

Class Management

- Viewing assigned classes
- Viewing class rosters
- Accessing student information

Attendance Recording

- Recording daily attendance for students
- Marking students as present, absent, late, or excused
- Adding remarks where necessary

Academic Assessments

- Entering student marks for tests and exams
- Adding comments on student performance
- Monitoring subject performance trends

Assignments and Homework

- Creating homework assignments
- Creating projects or revision tasks
- Setting submission deadlines
- Reviewing and grading student submissions
- Providing feedback to students

Learning Content

- Uploading recorded lessons

- Uploading educational materials
- Providing supporting documents for students

Practice Exams

- Creating practice tests
- Creating quizzes and exercises
- Reviewing quiz performance

Teachers are the primary academic contributors to the system.

Class Teacher Role

A **Class Teacher** is a teacher assigned to be responsible for the overall management of a specific class.

Each class must have **one designated class teacher**.

Example:

Class	Class Teacher
Grade 3A	Mr. Dube
Grade 3B	Mrs. Ncube
Form 2A	Mr. Moyo

The class teacher acts as the **academic supervisor for that class**.

Responsibilities of the Class Teacher

The class teacher has responsibilities that go beyond subject teaching.

1. Student Monitoring

The class teacher can:

- View all students in the class
- Monitor student performance across all subjects
- Monitor attendance records
- Identify struggling students

This allows the teacher to understand the **overall progress of each learner**.

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2. Attendance Oversight

While subject teachers may record attendance during lessons, the class teacher has authority to:

- Review attendance records
- Identify frequent absences
- Monitor attendance summaries for the entire class

This ensures proper student participation tracking.

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3. Academic Performance Review

The class teacher can view:

- All subject marks entered by teachers
- Student averages
- Class averages
- Student rankings

This allows them to monitor **overall class performance**.

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4. Final Report Comment

One of the most important roles of the class teacher is writing the **final comment on the student report card**.

The report contains two types of comments:

Subject Teacher Comment

Each subject teacher may write a comment related to the student's performance in that subject.

Example:

"John shows strong problem-solving skills but needs to practice algebra."

Class Teacher Final Comment

The class teacher writes the **overall evaluation of the student**.

Example:

"John has shown steady improvement throughout the term and should continue working hard to maintain progress."

This comment appears at the **bottom of the report card**.

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5. Report Approval

The class teacher is responsible for reviewing the results before they are finalized.

Responsibilities include:

- Reviewing subject marks entered by teachers
- Ensuring all subjects have marks
- Reviewing calculated averages
- Adding final student comments
- Approving reports for publishing

This ensures academic reports are verified before release.

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6. Promotion Recommendation

The class teacher may also recommend:

- Promotion to the next class
- Academic support for struggling students
- Special academic attention

This recommendation may appear in the final report.

Example:

"Promoted to Grade 4."

or

"Requires additional support in Mathematics."

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7. Class Management Features

The class teacher dashboard should include:

Class Overview

- List of students
- Class attendance summary
- Class average performance

Student Monitoring

- Student performance across subjects
- Student attendance history
- Assignment completion

Reports

- Review report cards
- Add final comments
- Approve reports

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8. Relationship with Other Roles

The class teacher works between several other roles.

Teachers

Subject teachers provide marks and subject comments.

HOD (Head of Department)

HODs oversee subject performance across departments.

School Administrator

The school administrator manages the academic structure and publishes reports.

Parents and Students

Parents and students ultimately receive the finalized reports.

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9. Class Teacher Workflow

The class teacher workflow during a school term typically follows this process:

1. Teachers teach lessons and record attendance.

2. Teachers record marks for assessments.
3. Continuous assessments accumulate during the term.
4. Exams are conducted.
5. The system calculates subject results.
6. Subject teachers add comments.
7. The class teacher reviews the class results.
8. The class teacher adds final comments for each student.
9. The class teacher approves the reports.
10. The school administrator publishes the reports.
11. Students and parents access the reports.

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10. Importance of the Class Teacher Role

The class teacher serves as the **primary academic guardian of the class**.

Their role ensures:

- Student progress is monitored holistically
- Attendance problems are identified
- Academic results are verified
- Students receive meaningful feedback

This role reflects how schools traditionally operate and ensures the digital system mirrors real-world educational structures.

2.5 Student Portal

Students use the platform primarily for learning and monitoring their academic progress.

Features available to students include:

Learning Access

- Viewing recorded lessons
- Accessing course materials
- Viewing subject resources

Assignments

- Viewing assigned homework
- Submitting homework
- Viewing teacher feedback

Practice Exams

- Taking practice quizzes
- Reviewing quiz scores
- Monitoring learning progress

Academic Reports

- Viewing report cards
- Viewing subject marks
- Viewing grades and averages

Attendance Monitoring

- Viewing attendance summaries

Students are able to track their learning progress throughout the academic term.

2.6 Parent Portal

Parents are able to monitor their children's academic progress and school participation.

A parent account may be linked to **multiple students**.

Parents can:

- View report cards
- Monitor student attendance
- Track homework completion
- View learning materials
- Monitor academic performance trends
- Receive announcements and notifications from the school

This portal provides transparency between schools and families.

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3. Academic Structure

The system must support the Zimbabwean education system from early childhood to advanced secondary levels.

The academic structure includes:

- Early Childhood Development (ECD)
- Primary School Grades

- Secondary School Forms

Schools may have **multiple classes per grade level**, often referred to as streams.

Example:

- Grade 3A
- Grade 3B
- Grade 3C

Each class has a **designated class teacher** responsible for overall student supervision and reporting.

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4. Teaching Structure

Teachers may be assigned to:

- Multiple subjects
- Multiple classes

For example, a teacher may teach:

- Mathematics to Grade 3A
- Mathematics to Grade 3B
- Science to Grade 4A

This flexible assignment structure ensures the system supports realistic school staffing models.

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5. Assessment System

The system supports a flexible academic assessment structure that reflects real school evaluation processes.

Schools may create multiple types of assessments including:

- Monthly tests
- Exercises
- Continuous assessments
- Mid-term examinations
- Final examinations

Assessments are organized into categories to support **continuous assessment and formal examinations**.

Continuous assessments contribute toward the final subject mark.

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6. Continuous Assessment and Final Exams

Student subject performance is calculated using two major components:

Continuous Assessment

Continuous assessment includes multiple smaller evaluations such as:

- Monthly tests
- Exercises
- Homework assessments
- Practice tests

These may be averaged or calculated using a median score depending on school policy.

Final Examination

Final examinations represent larger, more formal tests administered at the end of a learning period.

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7. Result Calculation

The final subject result is typically calculated using weighted components such as:

- Continuous assessment score
- Final examination score

The system automatically calculates:

- Final subject marks
- Subject grades
- Total marks
- Student averages

These results are used to generate report cards.

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8. Report Card Generation

Report cards summarize a student's academic performance for a given academic period.

Each report card includes:

Academic Results

- Subject marks
- Continuous assessment score
- Final exam score
- Final subject mark
- Grade

Attendance Summary

- Total days attended
- Total absences
- Late records

Teacher Comments

Each subject teacher may add comments describing the student's performance in that subject.

Class Teacher Final Comment

The class teacher adds a final comment summarizing the student's overall performance and progress.

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9. Attendance System

The system records student attendance on a daily basis.

Teachers can record attendance during class sessions.

Attendance records include:

- Present
- Absent
- Late
- Excused

Attendance summaries are displayed in report cards and student dashboards.

This allows schools and parents to track student participation and discipline.

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10. Homework and Assignments

Teachers can create assignments for students within their classes.

Assignments may include:

- Homework tasks
- Projects
- Exercises
- Revision tasks

Assignments include:

- Instructions
- Due dates
- Maximum scores
- Submission formats

Students can submit assignments digitally through the system.

Teachers review submissions and provide scores and feedback.

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11. Recorded Lessons and Learning Materials

Teachers may upload learning content to support student learning.

Content may include:

- Video lessons
- Audio recordings
- Documents
- Presentation slides

These resources allow students to revisit lessons outside the classroom.

Recorded lessons form part of the system's **E-Learning capability**.

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12. Practice Exams and Quizzes

Teachers may create practice exams that students can take for revision.

Practice exams may contain different question types including:

- Multiple choice questions

- True or false questions
- Short answer questions

Practice exams may be automatically graded where possible.

Results help students identify areas needing improvement.

Practice exams may optionally contribute to continuous assessment depending on school configuration.

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13. Student Performance Analytics

The system provides analytical insights into academic performance including:

- Student averages
- Class averages
- Subject pass rates
- Department performance
- Student ranking within classes

These analytics help school leadership monitor academic outcomes.

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14. Learning Workflow

A typical academic workflow operates as follows:

1. Teachers deliver lessons and upload learning materials.
2. Teachers assign homework or exercises.
3. Students complete assignments and take quizzes.
4. Teachers record attendance and marks.
5. Continuous assessments accumulate during the term.
6. Schools conduct formal examinations.
7. The system calculates results automatically.
8. Teachers add subject comments.
9. Class teachers add final comments.
10. School administrators publish report cards.
11. Students and parents view results through their portals.

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15. System Architecture Layers

The system can be understood in three functional layers.

Learning Layer

Responsible for delivering educational content.

Includes:

- Recorded lessons
- Learning materials
- Homework
- Practice exams

Assessment Layer

Responsible for evaluating student performance.

Includes:

- Tests
- Exams
- Continuous assessments
- Mark recording

Reporting Layer

Responsible for academic reporting and analytics.

Includes:

- Report cards
- Attendance summaries
- Teacher comments
- Academic performance analytics

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16. Multi-Tenant Platform Design

The system is built as a **multi-tenant platform** where multiple schools operate within the same infrastructure.

Each school maintains its own:

- students
- teachers
- classes
- subjects
- academic records

Schools cannot access each other's data.

This design enables the platform to scale to support many schools.

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17. System Goals

The primary goals of the system are:

- Digitize school administration
- Support modern e-learning
- Improve academic monitoring
- Provide transparent student performance tracking
- Enable communication between teachers, students, and parents
- Provide scalable infrastructure for multiple schools

If you want, the **next step** can be creating a **complete system module map and feature roadmap**, which will show **every module required (about 80+ features)** so development can be structured in phases like a real production SaaS platform.